

VINODH BALAKRISHNA

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PROFESSIONAL SUMMARY

Embedded and Systems Software Engineer with a strong foundation in low-level programming, memory management, concurrency, and inter-process communication. Skilled in building efficient, reliable software on Linux through deep understanding of operating system internals and system-level debugging.

EDUCATION

University of Central Missouri

Masters - Computer Science

Lee's Summit, MO

Aug 2023 - May 2025

Related Coursework: Operating Systems, Algorithms, Computer Networking & Security, Compiler Design & Construction, Cryptographic

Algorithms and Protocols, Web Application Security, Cloud Security

Bharath Institute of Higher Education and Research

B.Tech - Computer Science Engineering

Chennai, India

Jul 2017 - May 2021

TECHNICAL SKILLS

Programming: C/C++, Python, Shell Scripting, Java, Rust, x86 Assembly

Core Competencies: Data Structures & Algorithms, Memory Management, Concurrency (Multi-threading/Processing), IPC

Operating Systems: GNU Linux/Unix Systems, Embedded Linux Firmware

Networking: TCP/IP, UDP, SNMP, Networking Protocols

Frameworks & Libraries: Yocto Project, RDK, GNU Readline, Jsoncpp, ASIO (non-Boost)

Tools: CMake, GDB, QEMU, OpenSSL, Wireshark, Jenkins, Git, Splunk, Vim

EXPERIENCE

Embedded Linux Engineer

Comcast Corporation

Chennai, India

Dec 2021 - July 2023

- Contributed to the Platform Security team within the TPX unit, focusing on hardening embedded Linux systems for large-scale CPE deployments.
- Leveraged the **Yocto Project** build system and RDK software stack to create and manage custom Linux firmware for over 1 million CPE devices, including set-top boxes, routers, and TVs.
- Enhanced security in the custom Restricted Shell by implementing command logging and blocking unauthorized commands, significantly improving SSH access security to CPE devices.
- Implemented a Tab Completion feature in the Restricted Shell using the **Readline library** and a multi-threaded approach, which increased shell efficiency by 40% and improved user interaction.
- Migrated SBOM functionality from Yocto Kirkstone to Dunfell, ensuring backward compatibility while enhancing supply chain security and maintaining 100% transparency in the software inventory.
- Enabled security feature ASLR for multiple RDKB devices by utilizing position-independent code, which significantly reduced memory-related security vulnerabilities.
- Renewed 5+ **X.509 PKI certificates** to maintain secure TLS communications across internal components, ensuring encrypted interactions within the system.
- Securely removed the video analytics module from RDKB devices, optimizing firmware performance, improving system efficiency, and reducing maintenance overhead.
- Debugged RDKV and RDKB devices running on **ARM** and **Atom** platforms, using serial port connections to analyze logs, identify issues, and provide fixes for improved system stability.

Programmer Analyst Trainee (Intern)

Cognizant Solutions

Chennai, India

Aug 2021 - Dec 2021

- Led a PoC to integrate **ShellCheck** for automated Shell Script error detection, improving the development and code review process.
- Addressed **CVE vulnerabilities** by applying security patches, contributing to improved software security.
- Transitioned standard C code to **Safe C library**, addressing memory-related vulnerabilities.

PERSONAL PROJECTS

Torrent CLI - BitTorrent Client | Source Code: https://github.com/v1n0dh/torrent_cli

- Built a BitTorrent client using modern C++23 features such as concepts, smart pointers, and multi-threading for high-performance concurrent downloads.
- Created a full Bencode encoder and decoder to serialize/deserialize **.torrent** metadata files, enabling seamless integration with BitTorrent protocol requirements.
- Designed a zero-copy memory-mapped I/O system using **mmap**, **std::memcpy**, **msync**, and RAII-based resource management for efficient disk writes.

- Developed core BitTorrent protocol features, including peer wire communication, tracker requests (HTTP), message parsing, and piece/block download strategies.
- Built a thread-safe message-passing system using custom `Shared_Queue` classes to manage download tasks and inter-thread communication.
- Verified data integrity through **SHA1** hash validation and reconstructed original files from downloaded pieces with byte-level accuracy.
- Leveraged **ASIO** library for scalable networking, handling multiple peer connections asynchronously with fine-grained control.
- Applied mutexes and futures to coordinate peer threads and ensure reliable shutdown of asynchronous operations.
- Utilized Wireshark to inspect peer connections and analyze BitTorrent message flows (e.g., piece requests/responses); employed Valgrind for memory diagnostics and strace to trace syscalls and debug low-level I/O behavior.

Custom Linux Shell (Vshell) | *Source Code:* <https://github.com/v1n0dh/vshell>

- Developed a custom shell in C++, inspired by a coding challenge at codingchallenges.fyi, closely resembling bash.
- Incorporated key shell features, including command execution, I/O redirection, and pipes, facilitating efficient user interaction with the system.
- Integrated tab completion for files and command history, enhancing user experience and productivity.

Real-Time System Metrics Statusbar | *Source Code:* <https://github.com/v1n0dh/dwmstatus>

- Created a custom status bar for the dwm window manager using C programming, displaying real-time system metrics such as network speeds, Wi-Fi network, CPU temperature, RAM usage, battery status, and clock.
- Gathered system statistics directly from low-level file structures like `/sys/class` for battery and network information, and used **X11 library** to dynamically update the status bar.
- Integrated network speed calculation by monitoring changes in `RX/TX` bytes and implemented Wi-Fi network detection using shell commands.
- Optimized system resource usage by writing efficient C code, providing critical information without relying on external tools or scripts.

CERTIFICATIONS

- Rust Essential Training by Barron Stone, LinkedIn Learnings - [Certificate Link](#)
- Linux Device Drivers by Kevin Dankwardt, LinkedIn Learnings - [Certificate Link](#)